

CE3420

Assignment

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Roll Number: CH22BTECH11018

Hello Colab, Matplotlib Basics

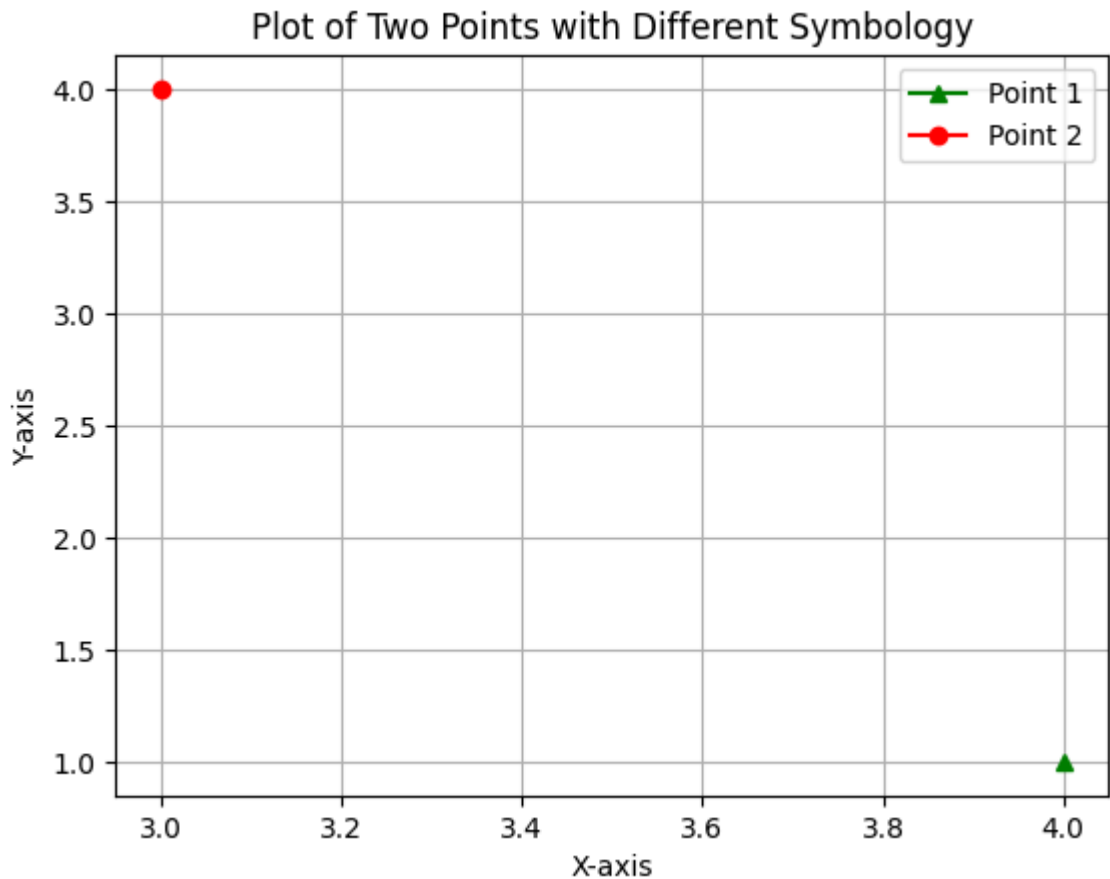
```
In [9]: """Question
Create a plot that displays the 2 given points with their x,y coordinates

point1: Plot it with green color and a triangle marker.
point2: Plot it with red color and a circle marker.
"""

import matplotlib.pyplot as plt

point1 = (4, 1)
point2 = (3, 4)

plt.plot(point1[0], point1[1], marker='^', color='green', label='Point 1')
plt.plot(point2[0], point2[1], marker='o', color='red', label='Point 2')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Plot of Two Points with Different Symbology')
plt.legend()
plt.grid()
plt.show()
```



Creating Charts, Creating Maps

```
In [10]: """Question
Plot the trend of Bicycle thefts as a line chart.
"""
```

```
Out[10]: 'Question\nPlot the trend of Bicycle thefts as a line chart.\n'
```

```
In [11]: import matplotlib.pyplot as plt
import os
import pandas as pd
import requests
```

```
In [12]: data_folder = 'Data'
output_folder = 'Assignment4'
```

```
In [13]: if not os.path.exists(data_folder):
os.mkdir(data_folder)
if not os.path.exists(output_folder):
os.mkdir(output_folder)

def download(url):
    filename = os.path.join(data_folder, os.path.basename(url))
    if not os.path.exists(filename):
        with requests.get(url, stream=True, allow_redirects=True) as r:
            with open(filename, 'wb') as f:
                for chunk in r.iter_content(chunk_size=8192):
                    f.write(chunk)
    print('Downloaded', filename)
```

```

files = [
    '2020-01-metropolitan-street.csv',
    '2020-02-metropolitan-street.csv',
    '2020-03-metropolitan-street.csv',
    '2020-04-metropolitan-street.csv',
    '2020-05-metropolitan-street.csv',
    '2020-06-metropolitan-street.csv',
    '2020-07-metropolitan-street.csv',
    '2020-08-metropolitan-street.csv',
    '2020-09-metropolitan-street.csv',
    '2020-10-metropolitan-street.csv',
    '2020-11-metropolitan-street.csv',
    '2020-12-metropolitan-street.csv'
]

```

In [14]: `data_url = 'https://github.com/spatialthoughts/python-dataviz-web/release/download/police.uk/'`

```

for f in files:
    url = os.path.join(data_url + f)
    download(url)

dataframe_list = []

for f in files:
    filepath = os.path.join(data_folder, f)
    df = pd.read_csv(filepath)
    dataframe_list.append(df)

```

Downloaded Data/2020-01-metropolitan-street.csv
 Downloaded Data/2020-02-metropolitan-street.csv
 Downloaded Data/2020-03-metropolitan-street.csv
 Downloaded Data/2020-04-metropolitan-street.csv
 Downloaded Data/2020-05-metropolitan-street.csv
 Downloaded Data/2020-06-metropolitan-street.csv
 Downloaded Data/2020-07-metropolitan-street.csv
 Downloaded Data/2020-08-metropolitan-street.csv
 Downloaded Data/2020-09-metropolitan-street.csv
 Downloaded Data/2020-10-metropolitan-street.csv
 Downloaded Data/2020-11-metropolitan-street.csv
 Downloaded Data/2020-12-metropolitan-street.csv

In [15]: `merged_df = pd.concat(dataframe_list)`
`merged_df`

Out [15]:

		Crime ID	Month	Reported by
0	24c2b43c34c7cd6ef847d69fbb042c9c730b1eebe95c52...		2020-01	Metropolitan Police Service
1		NaN	2020-01	Metropolitan Police Service
2		NaN	2020-01	Metropolitan Police Service
3		NaN	2020-01	Metropolitan Police Service
4		NaN	2020-01	Metropolitan Police Service
...		...	...	...
87931	05780af1c8515c41499ef396a33fc4e5c248533cecd9c4...		2020-12	Metropolitan Police Service
87932	54ce60d717f01752c5ff44f870e07a503e3fbeaae0b91e...		2020-12	Metropolitan Police Service
87933	580fdc2edfe26a09392ab6b9128732ba0c7cd05695387a...		2020-12	Metropolitan Police Service
87934	2730906a962972bed3589463c9be9dbb8ee8e594b779a3...		2020-12	Metropolitan Police Service
87935	826ceb8d3d2f79363e852449055994f5a8fec53b938fd7...		2020-12	Metropolitan Police Service

1181922 rows x 12 columns

```
In [16]: bicycle_thefts = merged_df[merged_df['Crime type'] == 'Bicycle theft']
bicycle_thefts['Month'] = pd.to_datetime(bicycle_thefts['Month'])
bicycle_thefts_count = bicycle_thefts.groupby('Month').size()

plt.figure(figsize=(10, 6))
plt.plot(bicycle_thefts_count.index, bicycle_thefts_count.values, marker=
plt.xlabel('Month'))
```

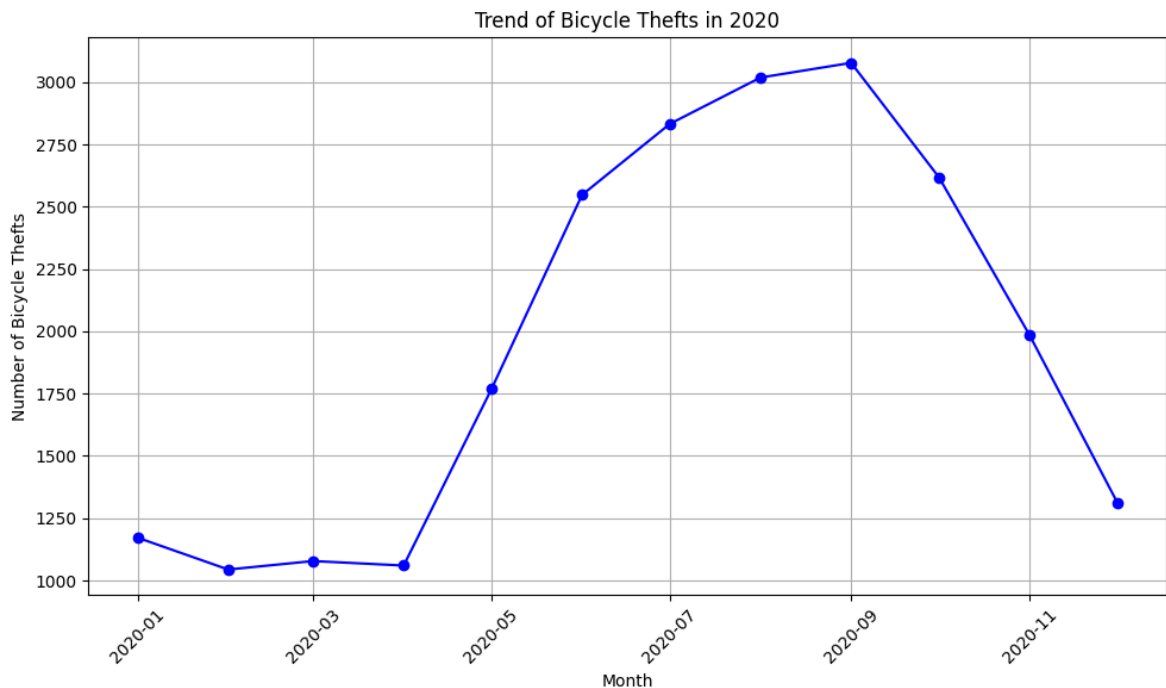
```
plt.ylabel('Number of Bicycle Thefts')
plt.title('Trend of Bicycle Thefts in 2020')
plt.xticks(rotation=45)
plt.grid()
plt.tight_layout()
plt.savefig(os.path.join(output_folder, 'bicycle_thefts_trend.png'))
plt.show()
```

/var/folders/2c/6ctx2yzx14s32xy8xmxjtdvw0000gn/T/ipykernel_13545/451095611.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
bicycle_thefts['Month'] = pd.to_datetime(bicycle_thefts['Month'])
```



In [17]: """Question
Plot the census tracts geodataframe tracts with just outlines and no fill
Display the map zoomed-in around the San Francisco area between Latitudes
"""

```
# %%capture
if 'google.colab' in str(get_ipython()):
    !pip install mapclassify
```

In [18]: `import` geopandas `as` gpd
`import` matplotlib.pyplot `as` plt
`import` os
`import` pandas `as` pd
`import` requests

In [19]: `data_folder = 'Data'`
`output_folder = 'Assignment4'`

`if not os.path.exists(data_folder):`
 `os.mkdir(data_folder)`

```
if not os.path.exists(output_folder):
    os.mkdir(output_folder)
```

```
In [20]: def download(url):
          filename = os.path.join(data_folder, os.path.basename(url))
          if not os.path.exists(filename):
              with requests.get(url, stream=True, allow_redirects=True) as r:
                  with open(filename, 'wb') as f:
                      for chunk in r.iter_content(chunk_size=8192):
                          f.write(chunk)
          print('Downloaded', filename)
```

```
In [21]: shapefile_name = 'tl_2019_06_tract'
          shapefile_exts = ['.shp', '.shx', '.dbf', '.prj']
          data_url = 'https://github.com/spatialthoughts/python-dataviz-web/releases/download/census/'

          for ext in shapefile_exts:
              url = data_url + shapefile_name + ext
              download(url)

          csv_name = 'ACSST5Y2019.S0101_data.csv'
          download(data_url + csv_name)
```

Downloaded Data/tl_2019_06_tract.shp
Downloaded Data/tl_2019_06_tract.shx
Downloaded Data/tl_2019_06_tract.dbf
Downloaded Data/tl_2019_06_tract.prj
Downloaded Data/ACSST5Y2019.S0101_data.csv

```
In [22]: shapefile_path = os.path.join(data_folder, shapefile_name + '.shp')
          tracts = gpd.read_file(shapefile_path)
          tracts
```

Out[22]:

	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME	NAMELSAD	MTFC
0	06	037	139301	06037139301	1393.01	Census Tract 1393.01	G502
1	06	037	139302	06037139302	1393.02	Census Tract 1393.02	G502
2	06	037	139502	06037139502	1395.02	Census Tract 1395.02	G502
3	06	037	139600	06037139600	1396	Census Tract 1396	G502
4	06	037	139701	06037139701	1397.01	Census Tract 1397.01	G502
...	...	...	...	...	...	...	.
8052	06	001	443001	06001443001	4430.01	Census Tract 4430.01	G502
8053	06	001	443002	06001443002	4430.02	Census Tract 4430.02	G502
8054	06	001	443102	06001443102	4431.02	Census Tract 4431.02	G502
8055	06	001	443301	06001443301	4433.01	Census Tract 4433.01	G502
8056	06	037	137501	06037137501	1375.01	Census Tract 1375.01	G502

8057 rows x 13 columns

```
In [23]: csv_path = os.path.join(data_folder, csv_name)
table = pd.read_csv(csv_path, skiprows=[1])
table
```


Out [23]:

	GEO_ID	NAME	S0101_C01_001E	S0101_C01_001M	S0101_C01_001H
0	1400000US06001400100	Census Tract 4001, Alameda County, California	3120	208	
1	1400000US06001400200	Census Tract 4002, Alameda County, California	2007	120	
2	1400000US06001400300	Census Tract 4003, Alameda County, California	5051	309	
3	1400000US06001400400	Census Tract 4004, Alameda County, California	4007	344	
4	1400000US06001400500	Census Tract 4005, Alameda County, California	4124	357	
...	...	...	...	...	...
8052	1400000US06115040800	Census Tract 408, Yuba County, California	4652	247	
8053	1400000US06115040901	Census Tract 409.01, Yuba County, California	2720	298	
8054	1400000US06115040902	Census Tract 409.02, Yuba County, California	1653	334	
8055	1400000US06115041000	Census Tract 410, Yuba County, California	7298	789	

		GEO_ID	NAME	S0101_C01_001E	S0101_C01_001M	S0101_C01_001H
			County, California			
			Census Tract			
8056	1400000US06115041100	411	Yuba County, California	4122	510	

8057 rows x 458 columns

```
In [24]: filtered = table[['GEO_ID', 'NAME', 'S0101_C01_001E']]
filtered = filtered.rename(columns = {'S0101_C01_001E': 'Population', 'GEO_ID': 'GEOID'})
filtered['GEOID'] = filtered.GEOID.str[-11:]

In [25]: gdf = tracts.merge(filtered, on='GEOID')
gdf
```

Out[25]:

	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME_x	NAMELSAD	MTFC
0	06	037	139301	06037139301	1393.01	Census Tract 1393.01	G50%
1	06	037	139302	06037139302	1393.02	Census Tract 1393.02	G50%
2	06	037	139502	06037139502	1395.02	Census Tract 1395.02	G50%
3	06	037	139600	06037139600	1396	Census Tract 1396	G50%
4	06	037	139701	06037139701	1397.01	Census Tract 1397.01	G50%
...	...	...	...	...	...	...	...
8052	06	001	443001	06001443001	4430.01	Census Tract 4430.01	G50%
8053	06	001	443002	06001443002	4430.02	Census Tract 4430.02	G50%
8054	06	001	443102	06001443102	4431.02	Census Tract 4431.02	G50%

	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME_x	NAMELSAD	MTFC
8055	06	001	443301	06001443301	4433.01	Census Tract 4433.01	G50%
8056	06	037	137501	06037137501	1375.01	Census Tract 1375.01	G50%

8057 rows x 15 columns

```
In [26]: gdf['density'] = 1e6*gdf['Population']/gdf['ALAND']
gdf
```

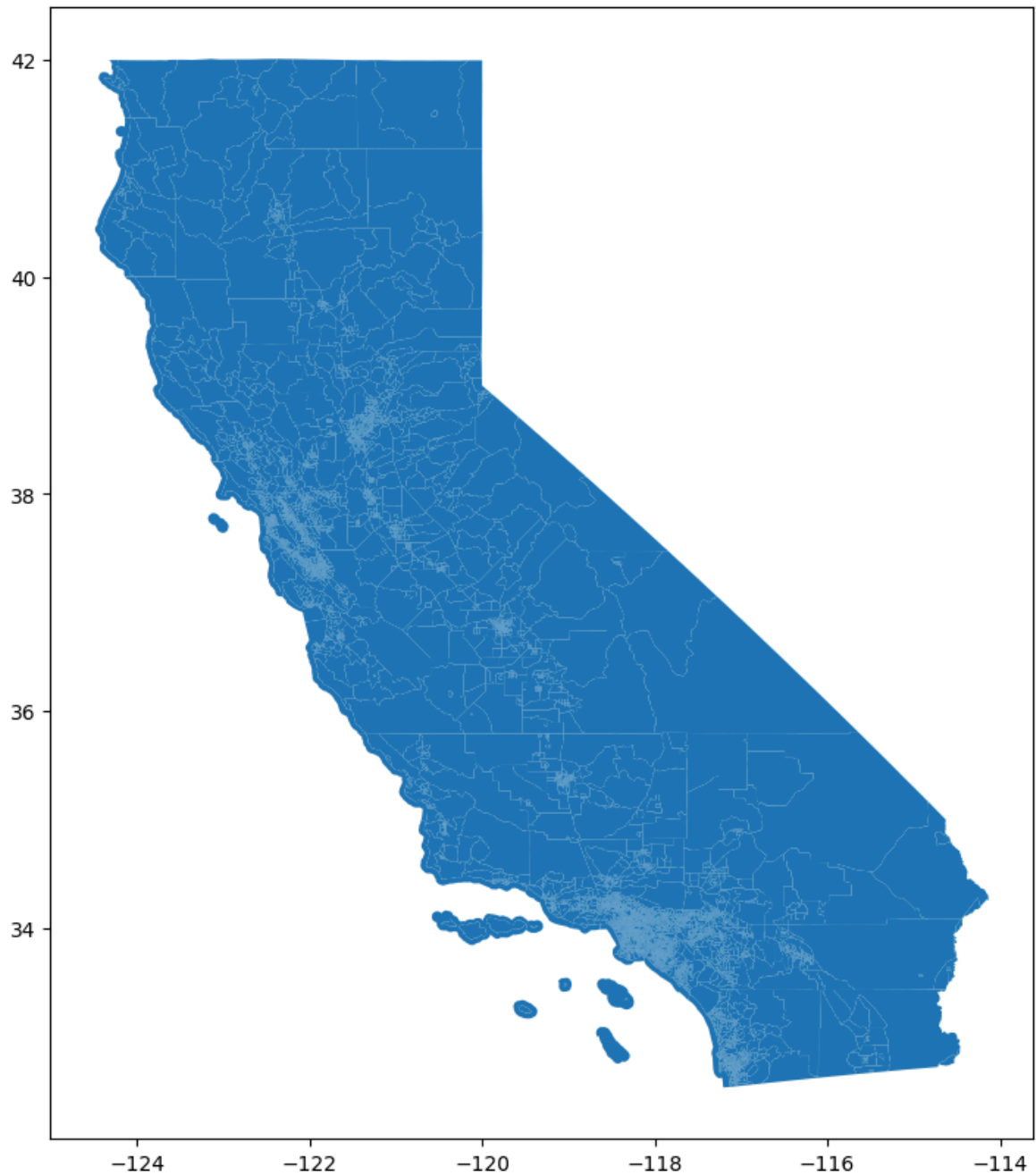
Out[26]:

	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME_x	NAMELSAD	MTFC
0	06	037	139301	06037139301	1393.01	Census Tract 1393.01	G50%
1	06	037	139302	06037139302	1393.02	Census Tract 1393.02	G50%
2	06	037	139502	06037139502	1395.02	Census Tract 1395.02	G50%
3	06	037	139600	06037139600	1396	Census Tract 1396	G50%
4	06	037	139701	06037139701	1397.01	Census Tract 1397.01	G50%
...	...	...	...	...	...	...	...
8052	06	001	443001	06001443001	4430.01	Census Tract 4430.01	G50%
8053	06	001	443002	06001443002	4430.02	Census Tract 4430.02	G50%
8054	06	001	443102	06001443102	4431.02	Census Tract 4431.02	G50%

	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME_x	NAMELSAD	MTFC
8055	06	001	443301	06001443301	4433.01	Census Tract 4433.01	G50%
8056	06	037	137501	06037137501	1375.01	Census Tract 1375.01	G50%

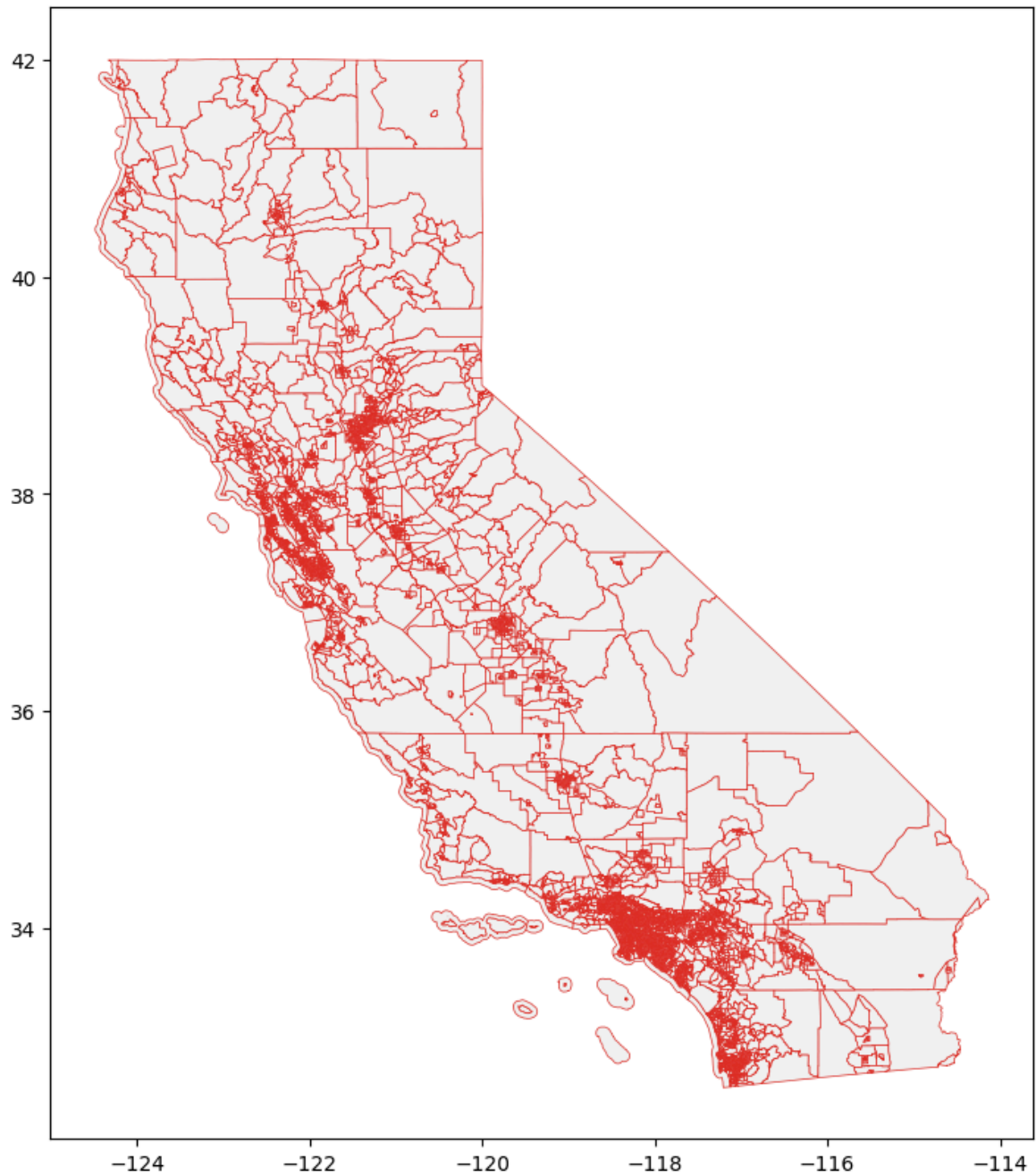
8057 rows x 16 columns

```
In [27]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax)
plt.show()
```



```
In [28]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax, facecolor='#f0f0f0', edgecolor='#de2d26', linewidth=0.5)

plt.show()
```



```
In [29]: %pip install mapclassify
fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax, column='density', cmap='RdYlGn_r', scheme='quantiles')
plt.show()
```


Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: mapclassify in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (2.8.1)

Requirement already satisfied: scipy>=1.8 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mapclassify) (1.13.1)

Requirement already satisfied: scikit-learn>=1.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mapclassify) (1.6.1)

Requirement already satisfied: pandas!=1.5.0,>=1.4 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mapclassify) (2.3.3)

Requirement already satisfied: networkx>=2.7 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mapclassify) (3.2.1)

Requirement already satisfied: numpy>=1.23 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mapclassify) (2.0.2)

Requirement already satisfied: python-dateutil>=2.8.2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pandas!=1.5.0,>=1.4->mapclassify) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pandas!=1.5.0,>=1.4->mapclassify) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pandas!=1.5.0,>=1.4->mapclassify) (2025.3)

Requirement already satisfied: six>=1.5 in /Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/site-packages (from python-dateutil>=2.8.2->pandas!=1.5.0,>=1.4->mapclassify) (1.15.0)

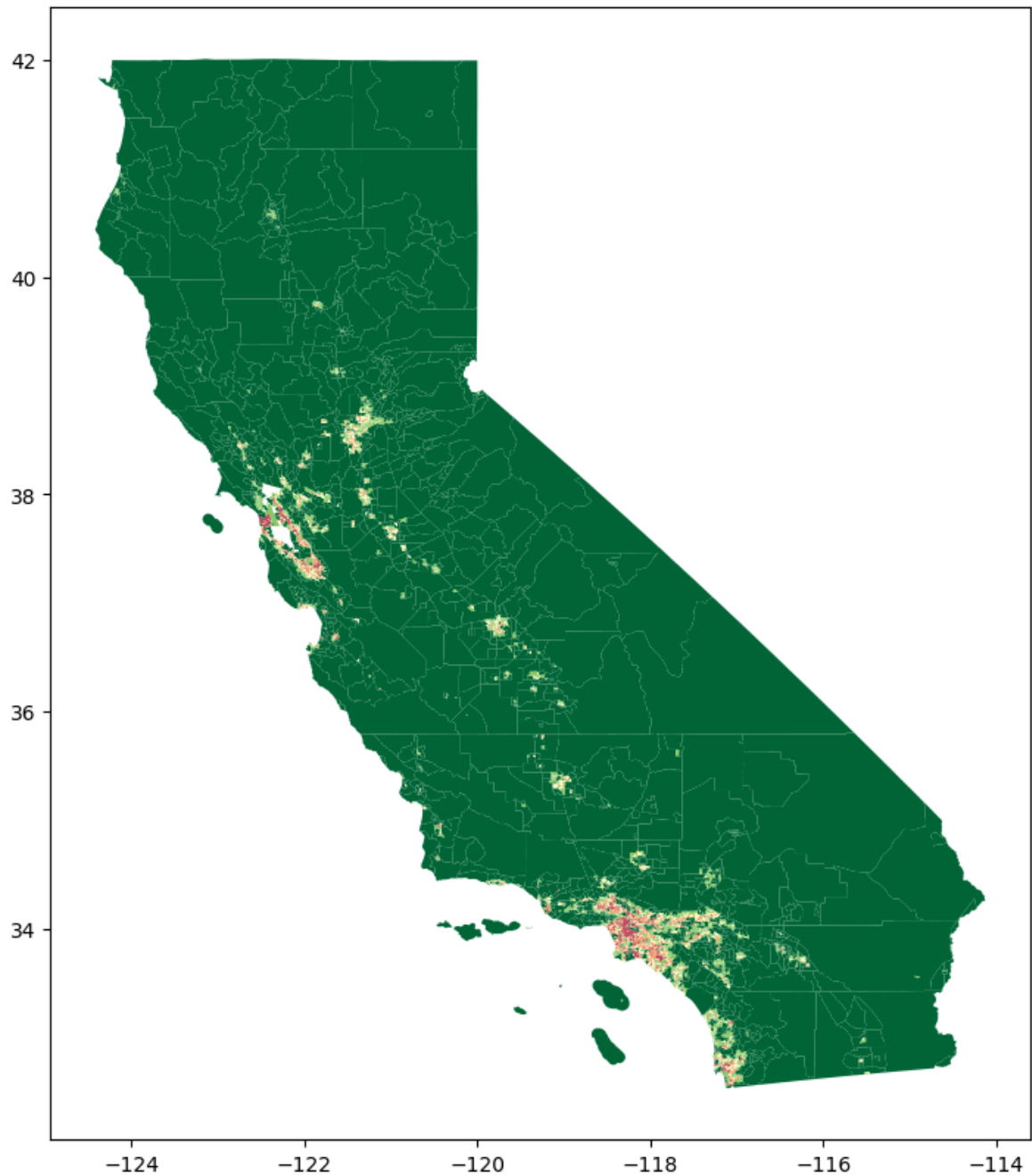
Requirement already satisfied: threadpoolctl>=3.1.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from scikit-learn>=1.0->mapclassify) (3.6.0)

Requirement already satisfied: joblib>=1.2.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from scikit-learn>=1.0->mapclassify) (1.5.3)

WARNING: You are using pip version 21.2.4; however, version 26.0.1 is available.

You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --upgrade pip' command.

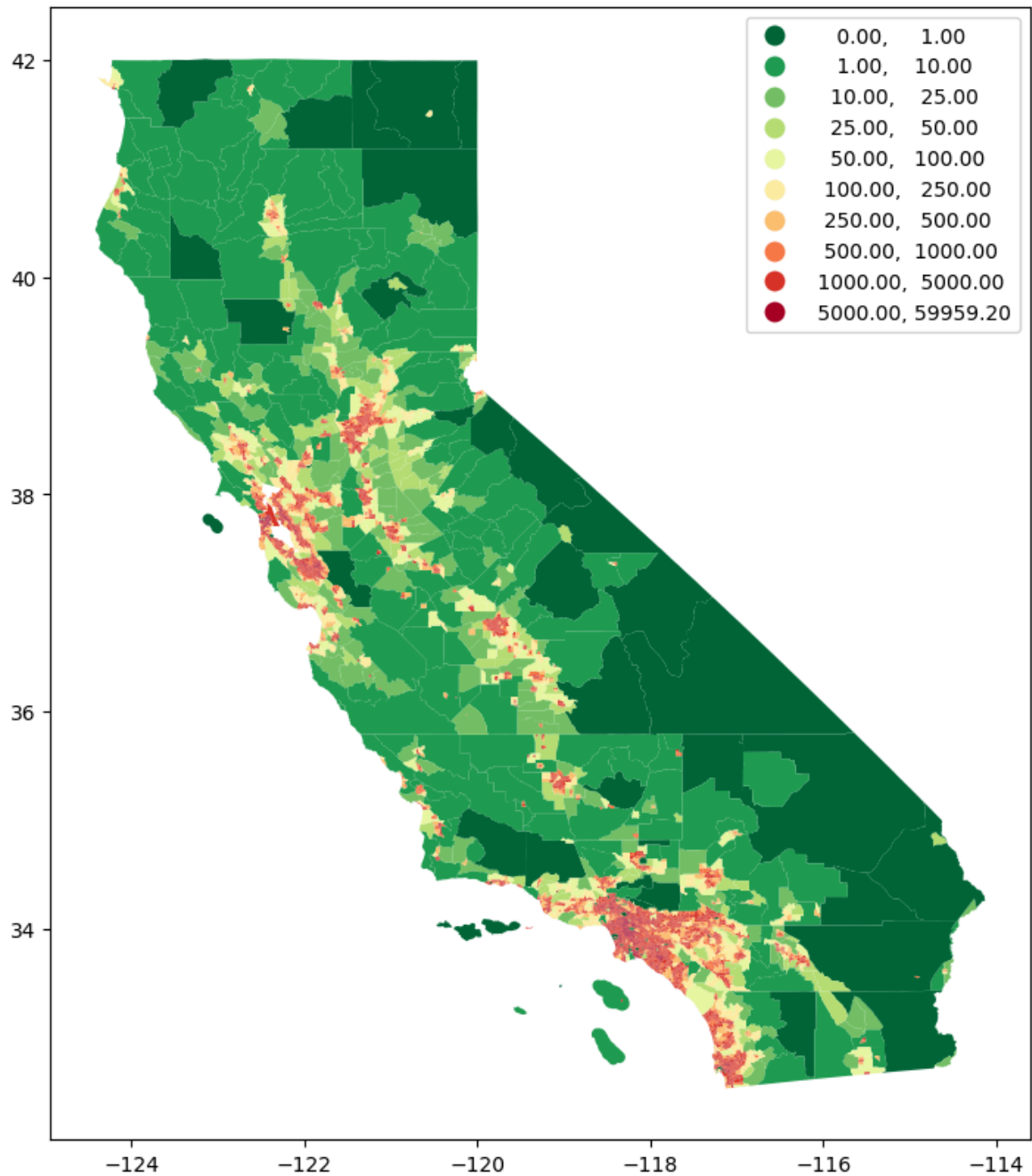
Note: you may need to restart the kernel to use updated packages.



```
In [30]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)

classification_kws={
    'bins': [1,10,25,50,100, 250, 500, 1000, 5000]
}
gdf.plot(ax=ax, column='density', cmap='RdYlGn_r', scheme='User_Defined',
         classification_kws=classification_kws, legend=True)

plt.show()
```



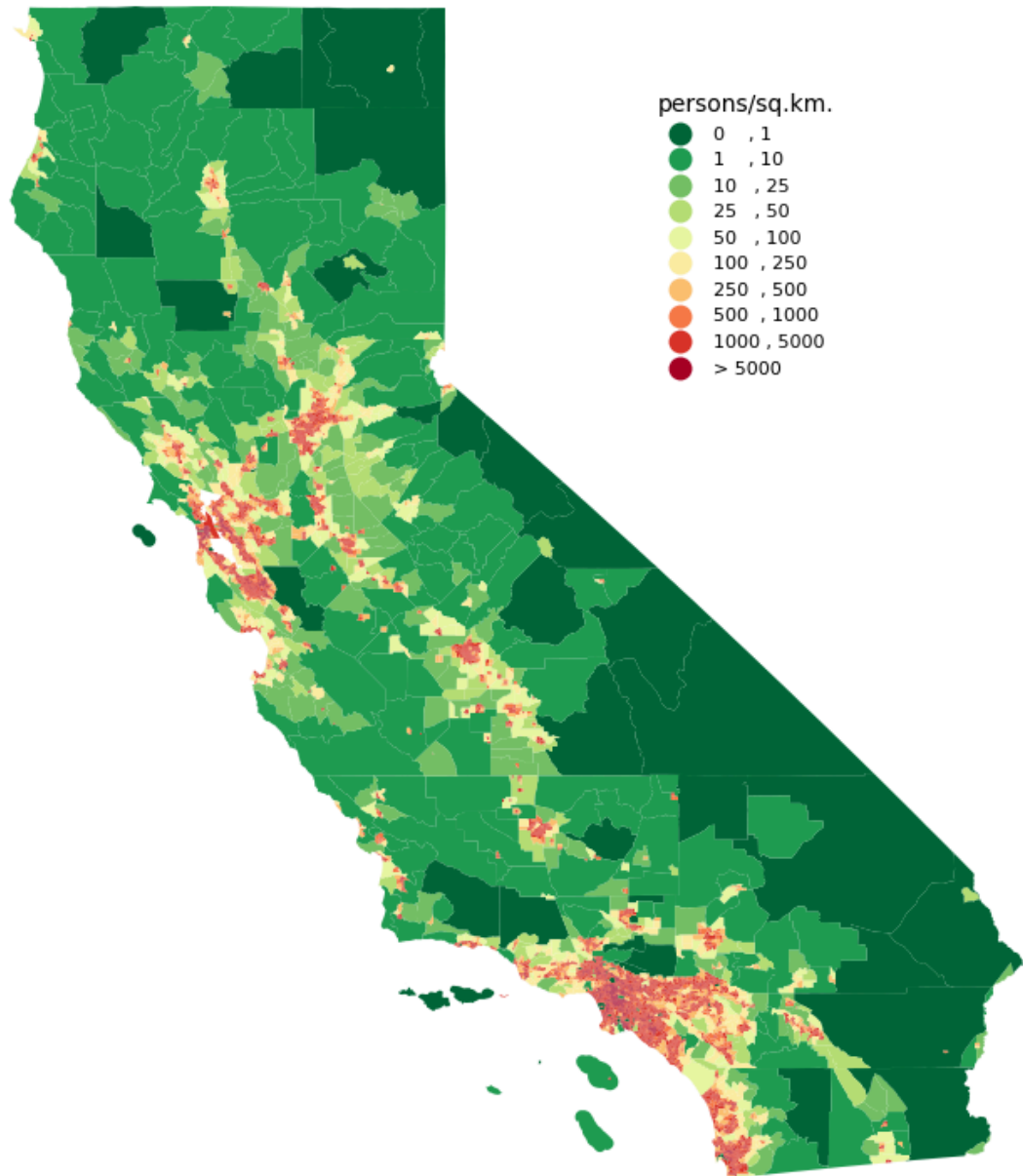
```
In [31]: legend_kwds= {
    'loc': 'upper right',
    'bbox_to_anchor': (0.8, 0.9),
    'fmt': '{:<5.0f}',
    'frameon': False,
    'fontsize': 8,
    'title': 'persons/sq.km.'
}
classification_kwds={
    'bins':[1,10,25,50,100, 250, 500, 1000, 5000]
}

fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax, column='density', cmap='RdYlGn_r', scheme='User_Defined',
         classification_kwds=classification_kwds,
         legend=True, legend_kwds=legend_kwds)

ax.set_axis_off()
```

```
# Change the last entry in the legend to '>5000'
legend = ax.get_legend()
legend.texts[-1].set_text('> 5000')

plt.show()
```



```
In [32]: legend_kws= {
    'loc': 'upper right',
    'bbox_to_anchor': (0.8, 0.9),
    'fmt': '{:<5.0f}',
    'frameon': False,
    'fontsize': 8,
    'title': 'persons/sq.km.'
}
classification_kws={
    'bins': [1,10,25,50,100, 250, 500, 1000, 5000]
}
```

```
fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax, column='density', cmap='RdYlGn_r', scheme='User_Defined',
         classification_kws=classification_kws,
         legend=True, legend_kws=legend_kws)

ax.set_axis_off()

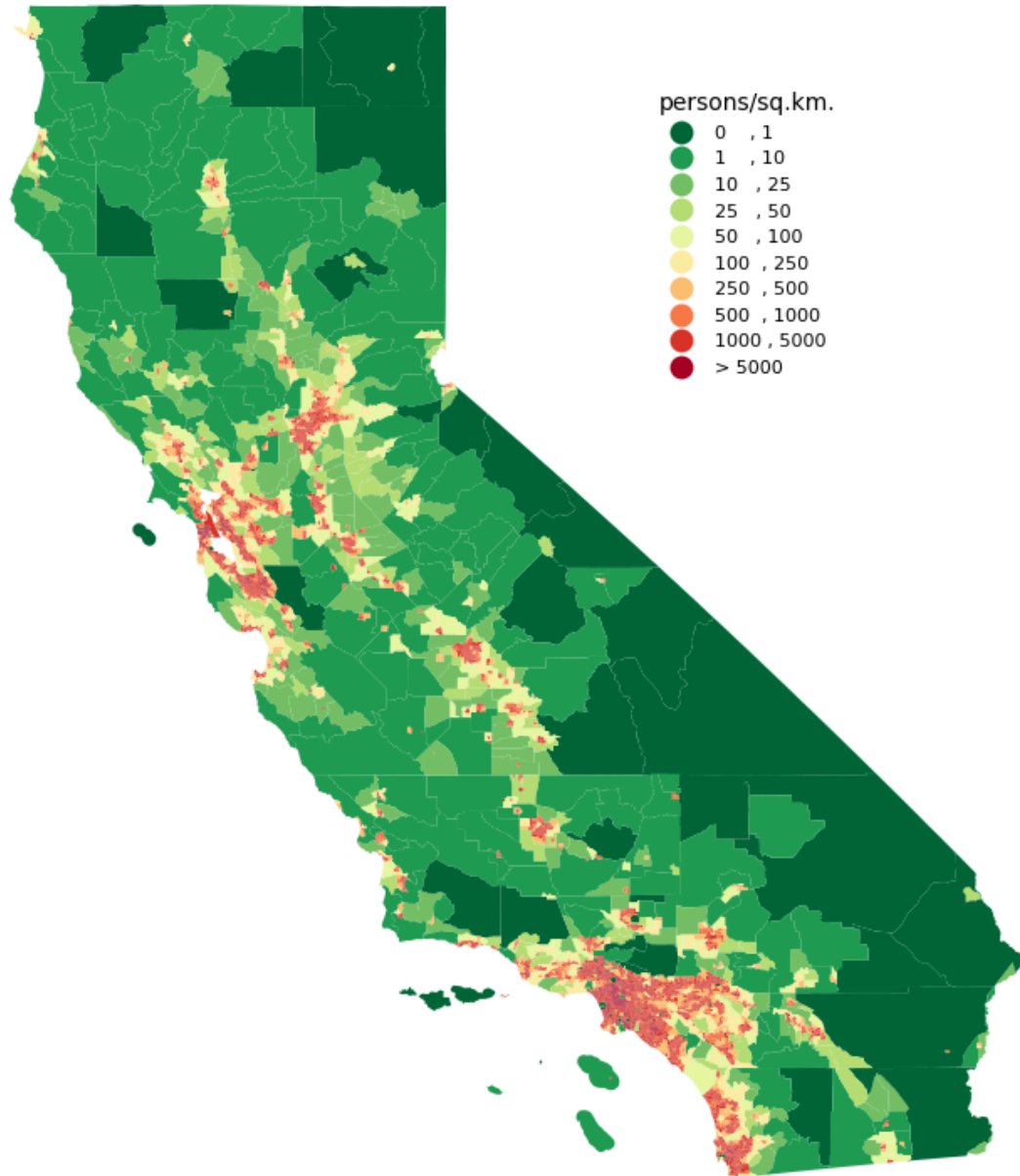
# Change the last entry in the legend to '>5000'
legend = ax.get_legend()
legend.texts[-1].set_text('> 5000')

# Add a title
ax.set_title('California Population Density (2019)', size = 18)

output_path = os.path.join(output_folder, 'california_pop.png')
plt.savefig(output_path, dpi=300)

plt.show()
```

California Population Density (2019)



In [33]:

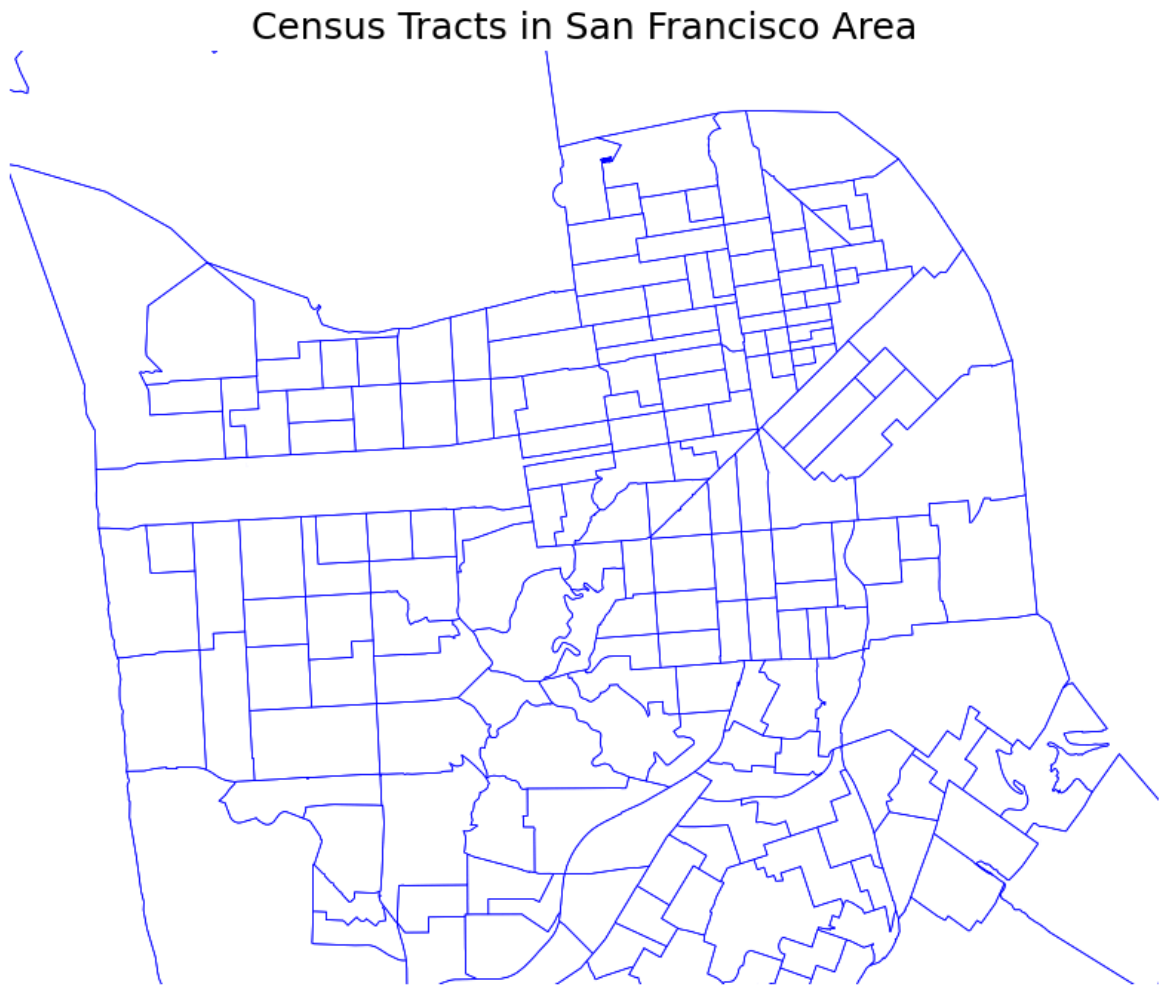
```
"""
Plot the census tracts geodataframe tracts with just outlines and no fill
Display the map zoomed-in around the San Francisco area between Latitudes

Hints:

Set the facecolor option to 'none' for no fills. Refer to the style_kwds
Use the set_xlim() and set_ylim() methods to set the view area of the Axes
"""

fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10,10)
gdf.plot(ax=ax, facecolor='none', edgecolor='blue', linewidth=0.5)
ax.set_xlim(-122.53, -122.36)
ax.set_ylim(37.71, 37.82)
ax.set_axis_off()
```

```
ax.set_title('Census Tracts in San Francisco Area', size=18)
plt.show()
```



Using Basemaps

```
In [34]: %%capture
if 'google.colab' in str(get_ipython()):
    !pip install contextily
```

```
In [36]: %pip install contextily
import contextily as cx
import geopandas as gpd
import matplotlib.pyplot as plt
import os
import requests
import shapely
```

Defaulting to user installation because normal site-packages is not writeable

Collecting contextily

Downloading contextily-1.6.2-py3-none-any.whl (17 kB)

Collecting mercantile

Downloading mercantile-1.2.1-py3-none-any.whl (14 kB)

Requirement already satisfied: rasterio in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (1.4.3)

Requirement already satisfied: geopy in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (2.4.1)

Requirement already satisfied: matplotlib in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (3.9.4)

Requirement already satisfied: requests in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (2.32.5)

Requirement already satisfied: pillow in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (11.3.0)

Requirement already satisfied: joblib in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from contextily) (1.5.3)

Collecting xyzservices

Downloading xyzservices-2025.11.0-py3-none-any.whl (93 kB)

|██| 93 kB 3.8 MB/s eta 0:00:01

Requirement already satisfied: geographiclib<3,>=1.52 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from geopy->contextily) (2.1)

Requirement already satisfied: packaging>=20.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (25.0)

Requirement already satisfied: cycler>=0.10 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (1.4.7)

Requirement already satisfied: importlib-resources>=3.2.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (6.5.2)

Requirement already satisfied: contourpy>=1.0.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (1.3.0)

Requirement already satisfied: python-dateutil>=2.7 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (2.9.0.post0)

Requirement already satisfied: numpy>=1.23 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (2.0.2)

Requirement already satisfied: pyparsing>=2.3.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (3.3.2)

Requirement already satisfied: fonttools>=4.22.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from matplotlib->contextily) (4.60.2)

Requirement already satisfied: zipp>=3.1.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from importlib-resources>=3.2.0->matplotlib->contextily) (3.23.0)

Requirement already satisfied: six>=1.5 in /Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/site-packages (from python-dateutil>=2.7->matplotlib->contextily) (1.15.0)

Requirement already satisfied: click>=3.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from mercantile->contextily) (8.1.8)

Requirement already satisfied: click-plugins in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (1.1.1.2)

Requirement already satisfied: affine in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (2.4.0)

Requirement already satisfied: attrs in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (25.4.0)

Requirement already satisfied: cligj>=0.5 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (0.7.2)

Requirement already satisfied: importlib-metadata in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (8.7.1)

Requirement already satisfied: certifi in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio->contextily) (2026.1.4)

Requirement already satisfied: charset-normalizer<4,>=2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests->contextily) (3.4.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests->contextily) (2.6.3)

Requirement already satisfied: idna<4,>=2.5 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests->contextily) (3.11)

Installing collected packages: xyzservices, mercantile, contextily

Successfully installed contextily-1.6.2 mercantile-1.2.1 xyzservices-2025.11.0

WARNING: You are using pip version 21.2.4; however, version 26.0.1 is available.

You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install --upgrade pip' command.

Note: you may need to restart the kernel to use updated packages.

```
In [37]: data_folder = 'Data'
output_folder = 'Assignment4'

os.makedirs(data_folder, exist_ok=True)
os.makedirs(output_folder, exist_ok=True)

def download(url):
    filename = os.path.join(data_folder, os.path.basename(url))
    if not os.path.exists(filename):
        with requests.get(url, stream=True, allow_redirects=True) as r:
            with open(filename, 'wb') as f:
                for chunk in r.iter_content(chunk_size=8192):
                    f.write(chunk)
        print('Downloaded', filename)

path_shapefile = 'upath17'
umbra_shapefile = 'umbra17'
penumbra_shapefile = 'penum17'

shapefile_exts = ['.shp', '.shx', '.dbf', '.prj']
data_url = 'https://github.com/spatialthoughts/python-dataviz-web/release/download/eclipse/'

for shapefile in [path_shapefile, umbra_shapefile, penumbra_shapefile]:
    for ext in shapefile_exts:
        url = data_url + shapefile + ext
        download(url)
```

Downloaded Data/upath17.shp
 Downloaded Data/upath17.shx
 Downloaded Data/upath17.dbf
 Downloaded Data/upath17.prj
 Downloaded Data/umbra17.shp
 Downloaded Data/umbra17.shx
 Downloaded Data/umbra17.dbf
 Downloaded Data/umbra17.prj
 Downloaded Data/penum17.shp
 Downloaded Data/penum17.shx
 Downloaded Data/penum17.dbf
 Downloaded Data/penum17.prj

```
In [38]: path_shapefile_path = os.path.join(
          data_folder, path_shapefile + '.shp')
path_gdf = gpd.read_file(path_shapefile_path)
path_gdf

umbra_shapefile_path = os.path.join(
          data_folder, umbra_shapefile + '.shp')
umbra_gdf = gpd.read_file(umbra_shapefile_path)
umbra_gdf[:5]
```

```
Out[38]:
```

	Name	Time	TZ	geometry
0	Umbra01	10:00	PDT	POLYGON ((-141.73242 44.78022, -141.7207 44.78...
1	Umbra02	10:10	PDT	POLYGON ((-130.59961 45.3565, -130.58789 45.35...
2	Umbra03	10:20	PDT	POLYGON ((-121.85742 45.20222, -121.8457 45.20...
3	Umbra04	11:30	MDT	POLYGON ((-114.7793 44.60543, -114.76758 44.60...
4	Umbra05	11:40	MDT	POLYGON ((-108.63867 43.72656, -108.62695 43.7...

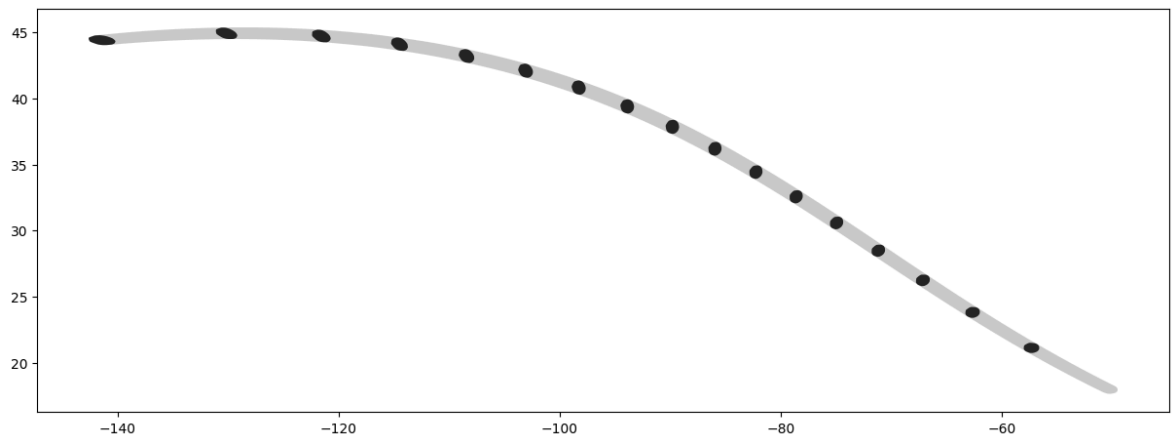
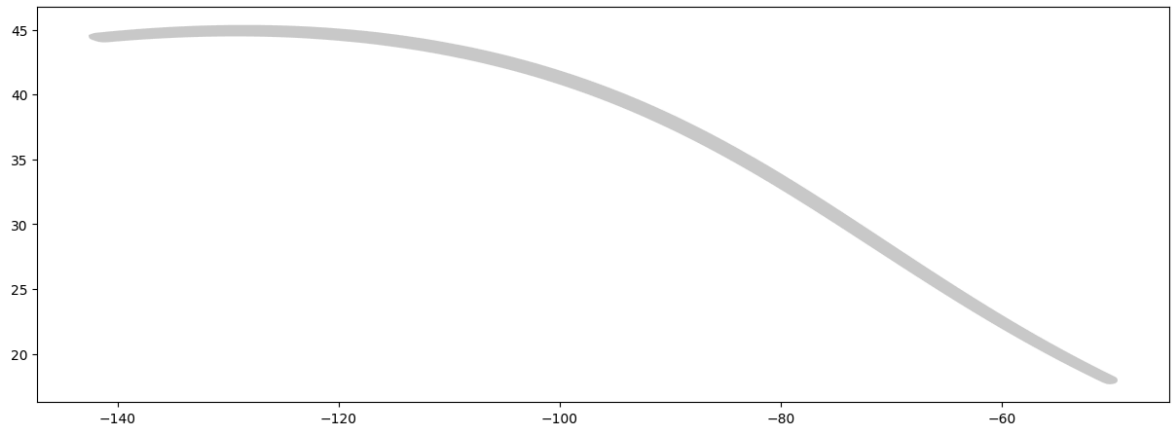
```
In [39]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(15,7)
path_gdf.plot(
    ax=ax,
    facecolor='#969696',
    edgecolor='none',
    alpha=0.5)
plt.show()

fig, ax = plt.subplots(1, 1)
fig.set_size_inches(15,7)
path_gdf.plot(
    ax=ax,
    facecolor='#969696',
    edgecolor='none',
    alpha=0.5)

umbra_gdf.plot(
    ax=ax,
    facecolor='#252525',
    edgecolor='none')

plt.show()
```

```
providers = cx.providers
providers
```



Out [39]: xyzservices.Bunch 41 items

- ▶ OpenStreetMap xyzservices.Bunch
- ▶ MapTilesAPI xyzservices.Bunch
- ▶ OpenSeaMap xyzservices.TileProvider
- ▶ OPNVKarte xyzservices.TileProvider
- ▶ OpenTopoMap xyzservices.TileProvider
- ▶ OpenRailwayMap xyzservices.TileProvider
- ▶ OpenFireMap xyzservices.TileProvider
- ▶ SafeCast xyzservices.TileProvider
- ▶ Stadia xyzservices.Bunch
- ▶ Thunderforest xyzservices.Bunch
- ▶ BaseMapDE xyzservices.Bunch
- ▶ CycloSM xyzservices.TileProvider
- ▶ Jawg xyzservices.Bunch
- ▶ MapBox xyzservices.TileProvider
- ▶ MapTiler xyzservices.Bunch
- ▶ TomTom xyzservices.Bunch
- ▶ Esri xyzservices.Bunch
- ▶ OpenWeatherMap xyzservices.Bunch
- ▶ HERE xyzservices.Bunch
- ▶ FreeMapSK xyzservices.TileProvider
- ▶ MtbMap xyzservices.TileProvider
- ▶ CartoDB xyzservices.Bunch
- ▶ HikeBike xyzservices.Bunch
- ▶ BasemapAT xyzservices.Bunch
- ▶ nlmaps xyzservices.Bunch
- ▶ NASAGIBS xyzservices.Bunch
- ▶ NLS xyzservices.Bunch
- ▶ JusticeMap xyzservices.Bunch
- ▶ GeoportailFrance xyzservices.Bunch
- ▶ OneMapSG xyzservices.Bunch
- ▶ USGS xyzservices.Bunch
- ▶ WaymarkedTrails xyzservices.Bunch
- ▶ OpenAIP xyzservices.TileProvider
- ▶ OpenSnowMap xyzservices.Bunch
- ▶ AzureMaps xyzservices.Bunch
- ▶ SwissFederalGeoportal xyzservices.Bunch
- ▶ TopPlusOpen xyzservices.Bunch
- ▶ Gaode xyzservices.Bunch
- ▶ Strava xyzservices.Bunch
- ▶ OrdnanceSurvey xyzservices.Bunch
- ▶ UN xyzservices.Bunch

```
In [40]: open_access_providers = providers.filter(requires_token=False)
open_access_providers
```

```
Out[40]: xyzservices.Bunch 30 items
```

- ▶ OpenStreetMap xyzservices.Bunch
- ▶ OpenSeaMap xyzservices.TileProvider
- ▶ OPNVKarte xyzservices.TileProvider
- ▶ OpenTopoMap xyzservices.TileProvider
- ▶ OpenRailwayMap xyzservices.TileProvider
- ▶ OpenFireMap xyzservices.TileProvider
- ▶ SafeCast xyzservices.TileProvider
- ▶ Stadia xyzservices.Bunch
- ▶ BaseMapDE xyzservices.Bunch
- ▶ CycloSM xyzservices.TileProvider
- ▶ Esri xyzservices.Bunch
- ▶ FreeMapSK xyzservices.TileProvider
- ▶ MtbMap xyzservices.TileProvider
- ▶ CartoDB xyzservices.Bunch
- ▶ HikeBike xyzservices.Bunch
- ▶ BasemapAT xyzservices.Bunch
- ▶ nlmaps xyzservices.Bunch
- ▶ NASAGIBS xyzservices.Bunch
- ▶ JusticeMap xyzservices.Bunch
- ▶ GeoportailFrance xyzservices.Bunch
- ▶ OneMapSG xyzservices.Bunch
- ▶ USGS xyzservices.Bunch
- ▶ WaymarkedTrails xyzservices.Bunch
- ▶ OpenAIP xyzservices.TileProvider
- ▶ OpenSnowMap xyzservices.Bunch
- ▶ SwissFederalGeoportal xyzservices.Bunch
- ▶ TopPlusOpen xyzservices.Bunch
- ▶ Gaode xyzservices.Bunch
- ▶ Strava xyzservices.Bunch
- ▶ UN xyzservices.Bunch

```
In [41]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(15,7)

path_gdf.plot(
    ax=ax,
    facecolor='#969696',
    edgecolor='none',
    alpha=0.5)

umbra_gdf.plot(
    ax=ax,
    facecolor='#252525',
```

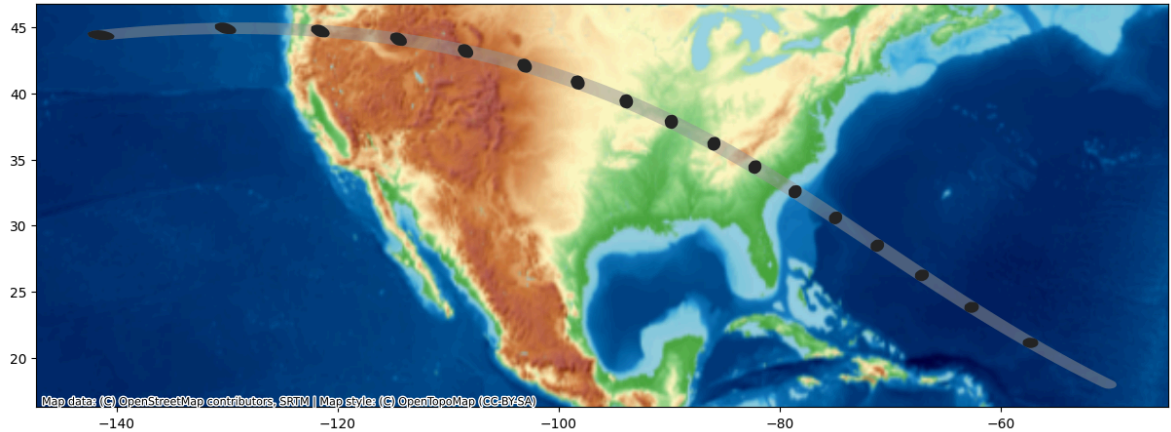
```

    edgecolor='none')

cx.add_basemap(
    ax,
    crs=path_gdf.crs,
    source=cx.providers.OpenTopoMap)

plt.show()

```



```

In [42]: zoom_level = cx.tile._calculate_zoom(*path_gdf.total_bounds)
         zoom_level

```

Out[42]: 3

```

In [43]: cx.howmany(*path_gdf.total_bounds, 4, ll=True)

```

Using zoom level 4, this will download 15 tiles

Out[43]: 15

```

In [44]: fig, ax = plt.subplots(1, 1)
         fig.set_size_inches(15,7)

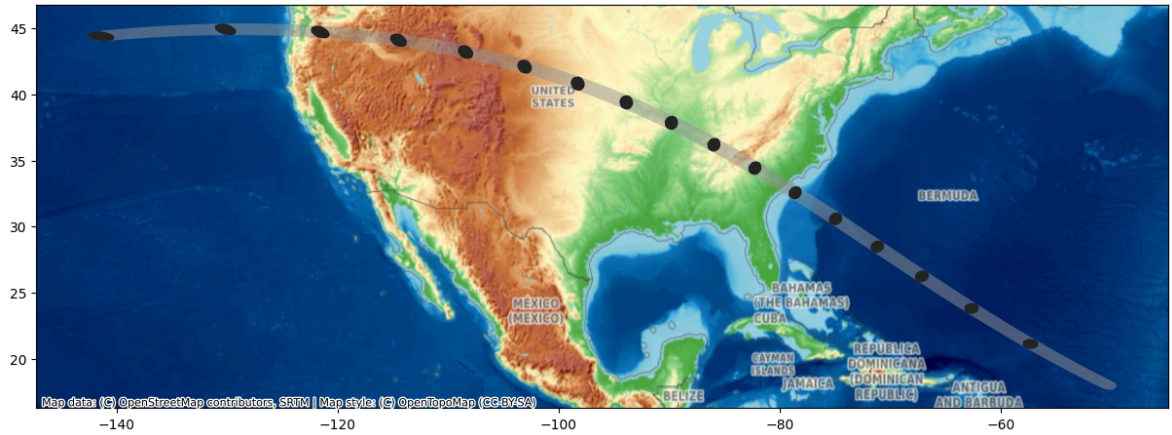
         path_gdf.plot(
             ax=ax,
             facecolor='#969696',
             edgecolor='none',
             alpha=0.5)

         umbra_gdf.plot(
             ax=ax,
             facecolor='#252525',
             edgecolor='none')

         cx.add_basemap(
             ax,
             crs=path_gdf.crs,
             source=cx.providers.OpenTopoMap,
             zoom=4)

         plt.show()

```



```
In [45]: crs = 'EPSG:9311'

path_reprojected = path_gdf.to_crs(crs)
umbra_reprojected = umbra_gdf.to_crs(crs)

# Use the bounding box coordinates for continental us
usa = shapely.geometry.box(-125, 24, -66, 49)
usa_gdf = gpd.GeoDataFrame(geometry=[usa], crs='EPSG:4326')
usa_gdf_reprojected = usa_gdf.to_crs(crs)
bounds = usa_gdf_reprojected.total_bounds
```

```
In [46]: fig, ax = plt.subplots(1, 1)
fig.set_size_inches(15,7)

# Set the bounds
ax.set_xlim(bounds[0], bounds[2])
ax.set_ylim(bounds[1], bounds[3])

path_reprojected.plot(
    ax=ax,
    facecolor='#969696',
    edgecolor='none',
    alpha=0.5)

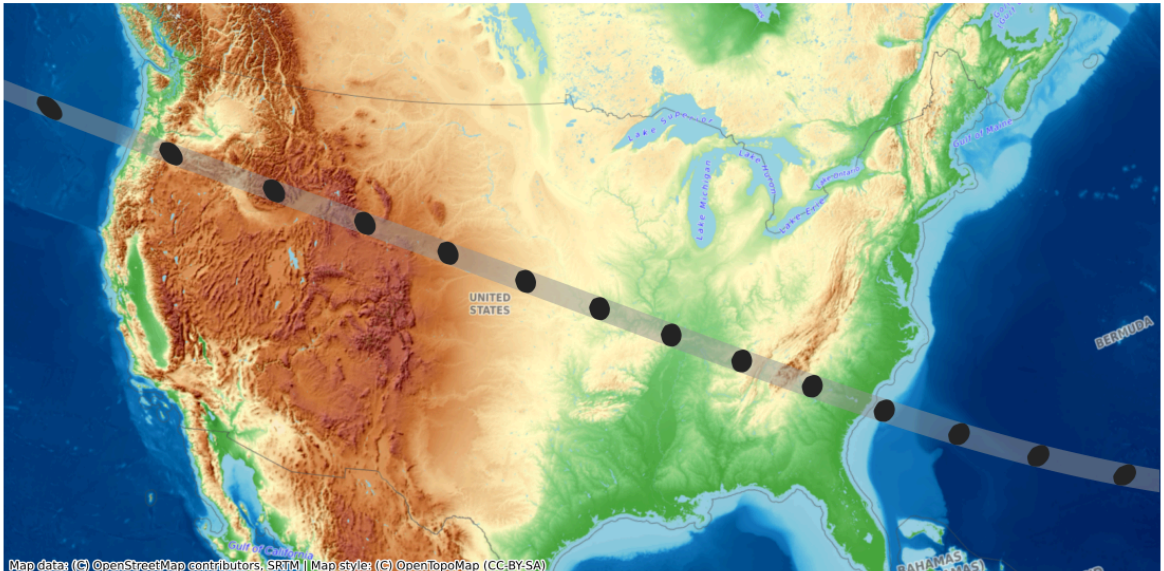
umbra_reprojected.plot(
    ax=ax,
    facecolor='#252525',
    edgecolor='none')

cx.add_basemap(
    ax,
    crs=path_reprojected.crs,
    source=cx.providers.OpenTopoMap,
    zoom=5)

ax.set_axis_off()
ax.set_title('2017 Total Solar Eclipse Path', size = 18)

plt.show()
```

2017 Total Solar Eclipse Path



```
In [48]: """Question
Our eclipse dataset also contains polygons for penumbra contours. Add the
Instead of the OpenTopoMap, create a visualization using another basemap.
"""

penumbra_shapefile_path = os.path.join(
    data_folder, penumbra_shapefile + '.shp')
penumbra_gdf = gpd.read_file(penumbra_shapefile_path)
penumbra_reprojected = penumbra_gdf.to_crs(crs)
penumbra_reprojected

# %pip install contextily
import contextily as ctx

fig, ax = plt.subplots(1, 1)
fig.set_size_inches(15, 7)

# Set the bounds
ax.set_xlim(bounds[0], bounds[2])
ax.set_ylim(bounds[1], bounds[3])

# Penumbra (20% transparency)
penumbra_reprojected.plot(
    ax=ax,
    column='Obscur',
    cmap='Greys',
    linewidth=1,
    alpha=0.2,
    legend=True
)

# Path on top
path_reprojected.plot(
    ax=ax,
    facecolor='none',
    edgecolor='red',
    linewidth=1.5
)

# Umbra on top
umbra_reprojected.plot(
```



```

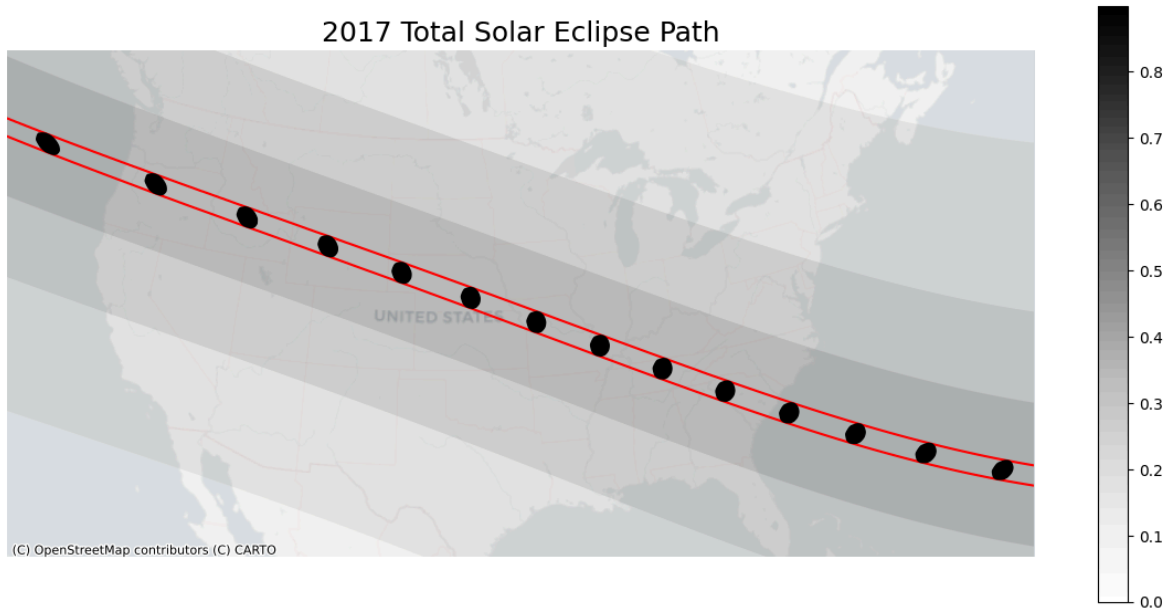
ax=ax,
facecolor='black',
edgecolor='black',
linewidth=1
)

# Basemap (pick one)
ctx.add_basemap(
    ax,
    crs=penumbra_reprojected.crs,
    source=ctx.providers.CartoDB.Positron
)

ax.set_axis_off()
ax.set_title('2017 Total Solar Eclipse Path',
size=18)

plt.show()

```



XArray Basics

In [49]: `"""Question`
`Display the median composite for the month of May.`

`The snippet below takes our time-series and aggregate it to a monthly med`
`"""`

Out[49]: 'Question\nDisplay the median composite for the month of May.\n\nThe snippet below takes our time-series and aggregate it to a monthly median composites groupby() method.\n'

In [50]: `import os`
`import matplotlib.pyplot as plt`
`%pip install pystac-client odc-stac rioxarray`
`import pystac_client`
`from odc.stac import stac_load`
`import xarray as xr`
`import rioxarray as rxr`

Defaulting to user installation because normal site-packages is not writeable

Collecting pystac-client
 Downloading pystac_client-0.8.3-py3-none-any.whl.metadata (5.2 kB)

Collecting odc-stac
 Downloading odc_stac-0.4.0-py3-none-any.whl.metadata (5.9 kB)

Collecting rioxarray
 Downloading rioxarray-0.15.0-py3-none-any.whl.metadata (6.5 kB)

Requirement already satisfied: requests>=2.28.2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pystac-client) (2.32.5)

Collecting pystac>=1.10.0 (from pystac[validation]>=1.10.0->pystac-client)
 Downloading pystac-1.10.1-py3-none-any.whl.metadata (6.4 kB)

Requirement already satisfied: python-dateutil>=2.8.2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pystac-client) (2.9.0.post0)

Requirement already satisfied: affine in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-stac) (2.4.0)

Collecting odc-geo>=0.4.7 (from odc-stac)
 Downloading odc_geo-0.4.10-py3-none-any.whl.metadata (6.1 kB)

INFO: pip is looking at multiple versions of odc-stac to determine which version is compatible with other requirements. This could take a while.

Collecting odc-stac
 Downloading odc_stac-0.3.11-py3-none-any.whl.metadata (5.2 kB)

Requirement already satisfied: rasterio!=1.3.0,!=1.3.1,>=1.0.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-stac) (1.4.3)

Collecting dask[array] (from odc-stac)
 Downloading dask-2024.8.0-py3-none-any.whl.metadata (3.8 kB)

Requirement already satisfied: numpy>=1.20.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-stac) (2.0.2)

Requirement already satisfied: pandas in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-stac) (2.3.3)

Collecting toolz (from odc-stac)
 Downloading toolz-1.1.0-py3-none-any.whl.metadata (5.1 kB)

Collecting xarray>=0.19 (from odc-stac)
 Downloading xarray-2024.7.0-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: packaging in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rioxarray) (25.0)

Requirement already satisfied: pyproj>=2.2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rioxarray) (3.6.1)

Requirement already satisfied: cachetools in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-geo>=0.4.7->odc-stac) (6.2.6)

Requirement already satisfied: shapely in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from odc-geo>=0.4.7->odc-stac) (2.0.7)

Requirement already satisfied: certifi in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pyproj>=2.2->rioxarray) (2026.1.4)

Requirement already satisfied: jsonschema~4.18 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pystac[validation]>=1.10.0->pystac-client) (4.25.1)

Requirement already satisfied: attrs>=22.2.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from jsonschema~4.18->pystac[validation]>=1.10.0->pystac-client) (25.4.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from jsonschema~4.18->pystac[validation]>=1.10.0->pystac-client) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from jsonschema~4.18->pystac[validation]>=1.10.0->pystac-client) (0.36.2)

Requirement already satisfied: rpds-py>=0.7.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from jsonschema~4.18->pystac[validation]>=1.10.0->pystac-client) (0.20.2)

rary/Python/3.9/lib/python/site-packages (from jsonschema~=4.18->pystac[validation]>=1.10.0->pystac-client) (0.27.1)
Requirement already satisfied: six>=1.5 in /Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/site-packages (from python-dateutil>=2.8.2->pystac-client) (1.15.0)
Requirement already satisfied: click>=4.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (8.1.8)
Requirement already satisfied: cligj>=0.5 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (0.7.2)
Requirement already satisfied: importlib-metadata in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (8.7.1)
Requirement already satisfied: click-plugins in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (1.1.1.2)
Requirement already satisfied: pyparsing in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (3.3.2)
Requirement already satisfied: typing-extensions>=4.4.0 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from referencing>=0.28.4->jsonschema~=4.18->pystac[validation]>=1.10.0->pystac-client) (4.15.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests>=2.28.2->pystac-client) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests>=2.28.2->pystac-client) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from requests>=2.28.2->pystac-client) (2.6.3)
Requirement already satisfied: pytz>=2020.1 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pandas->odc-stac) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from pandas->odc-stac) (2025.3)
Collecting cloudpickle>=1.5.0 (from dask[array]->odc-stac)
 Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting fsspec>=2021.09.0 (from dask[array]->odc-stac)
 Downloading fsspec-2025.10.0-py3-none-any.whl.metadata (10 kB)
Collecting partd>=1.4.0 (from dask[array]->odc-stac)
 Downloading partd-1.4.2-py3-none-any.whl.metadata (4.6 kB)
Collecting pyyaml>=5.3.1 (from dask[array]->odc-stac)
 Downloading pyyaml-6.0.3-cp39-cp39-macosx_11_0_arm64.whl.metadata (2.4 kB)
Requirement already satisfied: zipp>=3.20 in /Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages (from importlib-metadata->rasterio!=1.3.0,!1.3.1,>=1.0.0->odc-stac) (3.23.0)
Collecting locket (from partd>=1.4.0->dask[array]->odc-stac)
 Downloading locket-1.0.0-py2.py3-none-any.whl.metadata (2.8 kB)
Downloading pystac_client-0.8.3-py3-none-any.whl (33 kB)
Downloading odc_stac-0.3.11-py3-none-any.whl (89 kB)
Downloading pystac-1.10.1-py3-none-any.whl (182 kB)
Downloading rioxarray-0.15.0-py3-none-any.whl (53 kB)
Downloading odc_geo-0.4.10-py3-none-any.whl (155 kB)
Downloading xarray-2024.7.0-py3-none-any.whl (1.2 MB)
 1.2/1.2 MB 8.4 MB/s 0:00:00
Downloading dask-2024.8.0-py3-none-any.whl (1.2 MB)
 1.2/1.2 MB 9.5 MB/s 0:00:00
Downloading cloudpickle-3.1.2-py3-none-any.whl (22 kB)

Downloading fsspec-2025.10.0-py3-none-any.whl (200 kB)
Downloading partd-1.4.2-py3-none-any.whl (18 kB)
Downloading pyyaml-6.0.3-cp39-cp39-macosx_11_0_arm64.whl (174 kB)
Downloading toolz-1.1.0-py3-none-any.whl (58 kB)
Downloading locket-1.0.0-py2.py3-none-any.whl (4.4 kB)
Installing collected packages: toolz, pyyaml, locket, fsspec, cloudpickle, pystac, partd, odc-geo, xarray, dask, rioarray, pystac-client, odc-stac
Successfully installed cloudpickle-3.1.2 dask-2024.8.0 fsspec-2025.10.0 lo
cket-1.0.0 odc-geo-0.4.10 odc-stac-0.3.11 partd-1.4.2 pystac-1.10.1 pystac
-client-0.8.3 pyyaml-6.0.3 rioarray-0.15.0 toolz-1.1.0 xarray-2024.7.0
Note: you may need to restart the kernel to use updated packages.

```
In [51]: data_folder = "Data"
         output_folder = "Assignment4"

In [52]: os.makedirs(data_folder, exist_ok=True)
         os.makedirs(output_folder, exist_ok=True)

In [53]: latitude = 27.163
         longitude = 82.608
         year = 2023

In [54]: catalog = pystac_client.Client.open(
         "https://earth-search.aws.element84.com/v1"
         )

         # Create small bounding box (~1km radius)
         km2deg = 1.0 / 111
         x, y = (longitude, latitude)
         r = 1 * km2deg
         bbox = (x - r, y - r, x + r, y + r)

         # -----
         # Search Sentinel-2 Data
         # -----
         search = catalog.search(
             collections=["sentinel-2-c1-l2a"],
             bbox=bbox,
             datetime=f"{year}",
             query={"eo:cloud_cover": {"lt": 30}},
         )

         items = search.item_collection()
```

```
In [55]: # -----
         # Load data as XArray Dataset
         # -----
         ds = stac_load(
             items,
             bands=["red", "green", "blue", "nir"],
             resolution=10,
             bbox=bbox,
             chunks={}, # Use Dask
             groupby="solar_day",
         )

         # Compute (trigger Dask execution)
         ds = ds.compute()
```

```

# -----
# Create Monthly Median Composite
# -----
monthly = ds.groupby("time.month").median(dim="time")

# Convert Dataset to DataArray
monthly_da = monthly.to_array("band")

# -----
# Extract May (month=5)
# -----
may_median = monthly_da.sel(month=5)

```

```

In [56]: # -----
# Plot RGB Visualization
# -----
fig, ax = plt.subplots(1, 1)
fig.set_size_inches(6, 6)

may_median.sel(band=["red", "green", "blue"]).plot.imshow(
    ax=ax,
    robust=True
)

ax.set_title("May 2023 Median Composite (RGB)")
ax.set_axis_off()
ax.set_aspect("equal")

plt.show()

# -----
# Save as GeoTIFF
# -----
output_file_path = os.path.join(output_folder, "may_median_composite.tif")

may_median.rio.to_raster(output_file_path, compress="LZW")

print(f"Saved May median composite to: {output_file_path}")

```

May 2023 Median Composite (RGB)



Saved May median composite to: Assignment4/may_median_composite.tif

Mapping Gridded Datasets

```
In [ ]: """Question
Display the map in the Equal Earth projection.
"""
```

```
import os
import matplotlib.pyplot as plt
import xarray as xr
import cartopy.crs as ccrs
```

```
In [ ]: # -----
# Setup folders
# -----
data_folder = "data"
output_folder = "output"

os.makedirs(data_folder, exist_ok=True)
os.makedirs(output_folder, exist_ok=True)

# -----
# Download Dataset
# -----
def download(url):
    filename = os.path.join(data_folder, os.path.basename(url))
    if not os.path.exists(filename):
```

```

    from urllib.request import urlretrieve
    local, _ = urlretrieve(url, filename)
    print("Downloaded " + local)
    return filename

```

```

In [ ]: filename = "gistemp1200_GHCNV4_ERSSTv5.nc"

data_url = (
    "https://github.com/spatialthoughts/python-dataviz-web/releases/"
    "download/gistemp/"
)

file_path = download(data_url + filename)

```

```

In [ ]: # -----
# Load Data using XArray
# -----
ds = xr.open_dataset(file_path)
da = ds.tempanomaly

# -----
# Aggregate Monthly to Yearly Mean
# -----
yearly = da.groupby("time.year").mean(dim="time")

# -----
# Select Year
# -----
year = 2023
anomaly = yearly.sel(year=year)

# -----
# Plot using Equal Earth Projection (Exercise Solution)
# -----
projection = ccrs.EqualEarth()

```

```

In [ ]: cbar_kwargs = {
    "orientation": "horizontal",
    "fraction": 0.04,
    "pad": 0.05,
    "extend": "neither",
}

fig, ax = plt.subplots(
    1, 1,
    subplot_kw={"projection": projection}
)
fig.set_size_inches(10, 6)

anomaly.plot.imshow(
    ax=ax,
    vmin=-3,
    vmax=3,
    cmap="coolwarm",
    transform=ccrs.PlateCarree(),
    add_labels=False,
    cbar_kwargs=cbar_kwargs
)

```

```

ax.coastlines()
ax.set_title(f"Temperature Anomaly in {year} (°K)", fontsize=14)

output_path = os.path.join(output_folder, "equal_earth_anomaly.jpg")
plt.savefig(output_path, dpi=300)

plt.tight_layout()
plt.show()

print(f"Saved output map to: {output_path}")

```

Visualizing Rasters

```

In [57]: """Question
Add contours to the elevation plot below.
"""

```

```

Out[57]: 'Question\nAdd contours to the elevation plot below.\n'

```

```

In [58]: # srtm_with_contours.py

import os
import rioarray as rxr
from rioarray.merge import merge_arrays
import matplotlib.pyplot as plt

```

```

In [59]: # -----
# Setup folders
# -----
data_folder = "Data"
output_folder = "Assignment4"

os.makedirs(data_folder, exist_ok=True)
os.makedirs(output_folder, exist_ok=True)

# -----
# Download SRTM Tiles
# -----
def download(url):
    filename = os.path.join(data_folder, os.path.basename(url))
    if not os.path.exists(filename):
        from urllib.request import urlretrieve
        local, _ = urlretrieve(url, filename)
        print("Downloaded " + local)

srtm_tiles = [
    "N27E086.hgt",
    "N27E087.hgt",
    "N28E086.hgt",
    "N28E087.hgt"
]

data_url = (
    "https://github.com/spatialthoughts/python-dataviz-web/releases/"
    "download/srtm"
)

```



```

for tile in srtm_tiles:
    download(f"{data_url}/{tile}")

# -----
# Open and Merge Tiles
# -----
datasets = []

for tile in srtm_tiles:
    path = os.path.join(data_folder, tile)
    rds = rxr.open_rasterio(path)
    band = rds.sel(band=1)
    datasets.append(band)

merged = merge_arrays(datasets, method="first")

# -----
# Plot with Contours (Exercise Solution)
# -----
fig, ax = plt.subplots(1, 1)
fig.set_size_inches(10, 10)

# Base elevation image
merged.plot.imshow(
    ax=ax,
    cmap="viridis",
    add_labels=False,
    add_colorbar=False
)

# Add contours
merged.plot.contour(
    ax=ax,
    colors="white",
    levels=10
)

ax.set_axis_off()
plt.tight_layout()

# Save output
output_path = os.path.join(output_folder, "srtm_with_contours.png")
plt.savefig(output_path, dpi=300)

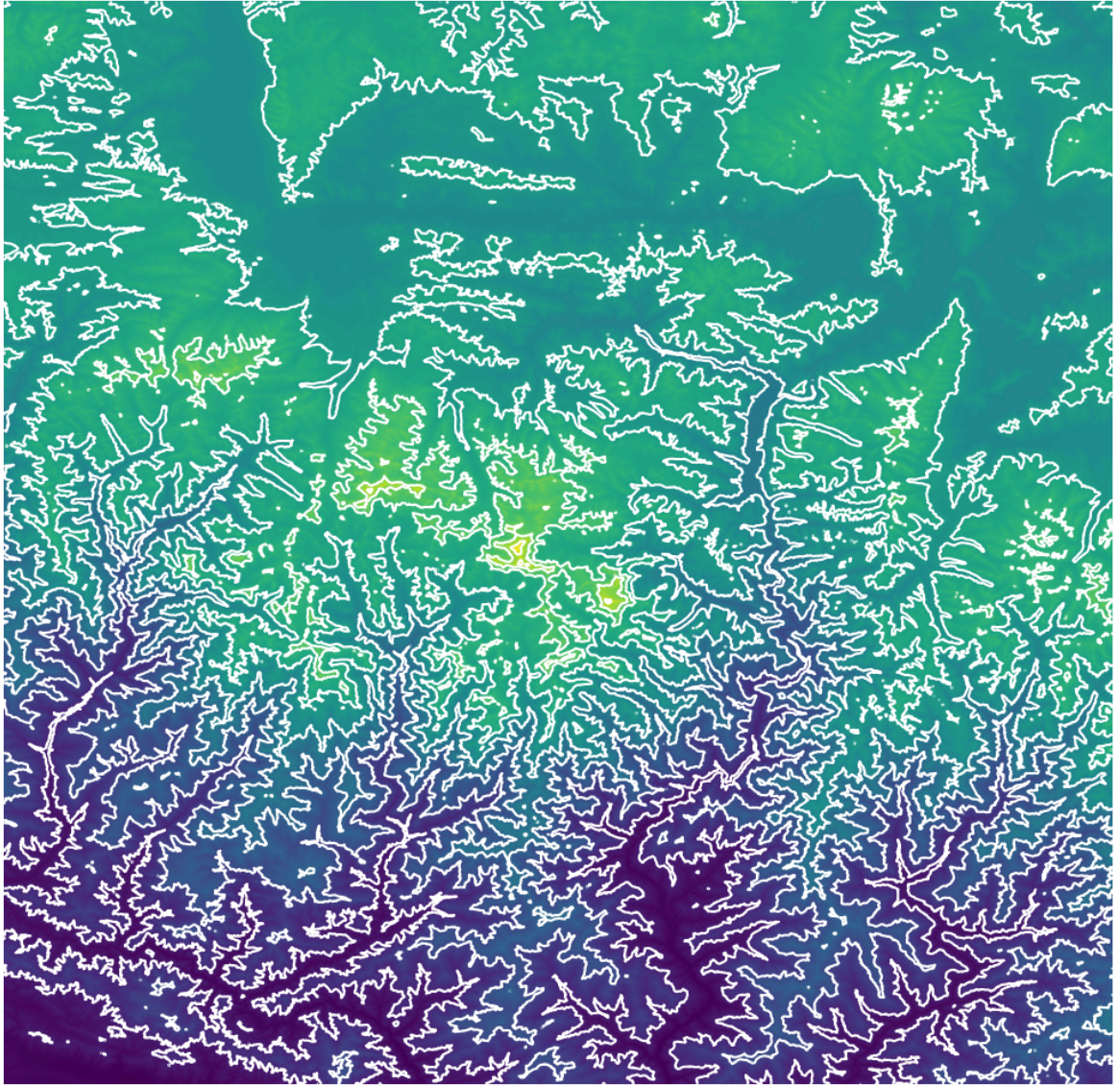
plt.show()

print(f"Saved contour map to: {output_path}")

```

Downloaded Data/N27E086.hgt
 Downloaded Data/N27E087.hgt
 Downloaded Data/N28E086.hgt
 Downloaded Data/N28E087.hgt

band = 1, spatial_ref = 0



Saved contour map to: Assignment4/srtm_with_contours.png